



# SFPPy Example 1: Mass Transfer from Monolayer Materials 🍏▶️🍎

## Synopsis

This example demonstrates **1D mass transfer** of **Irganox 1076** and **Irgafos 168** from a **100  $\mu\text{m}$  LDPE film** into a **fatty sandwich** over **10 days** at **7°C**.

💡 **Example 1** showcases functional syntax (low-level), providing direct and transparent access to simulation details. If you prefer higher-level syntax with less coding effort and fewer dependencies, check out **Example 3** and **Example 4**, which leverage operator overloading and advanced methods for a more streamlined experience.

## 🖥️🔬 SFPPy Example 1: Mass Transfer from Monolayer Materials 🍏▶️🍎

### Overview 📌

#### Simulation Steps 1 2 3 4

#### Expected Outcomes ✅

#### Dependencies 📦

#### Code Breakdown 📝

- 1 Define Contact Conditions
- 2 Define Sandwich Geometry 🥪
- 3 Define the Migrant (Irganox 1076) 🏭
- 4 Create the LDPE Film 📄
- 5 Define Food Properties 🍔
- 6 Run Migration Simulation 🚀
- 7 Compare Migration for Two Additives: Irganox 1076 ( `m1` ) and Irgafos 168 ( `m2` ) 📊
- 8 Save and Print Figures 📄
- 9 Results 📈

## Overview 📌

### Simulation Steps 1 2 3 4

#

1. Define the sandwich geometry (cylindrical shape).
2. Set up the polymer layer (LDPE with additives).
3. Define the food properties (real, semisolid, fatty).
4. Simulate mass transfer for:
  - Irganox 1076 in LDPE.
  - Irgafos 168 in LDPE.

5. Compare migration kinetics between both cases.
6. Save and print all simulation results.

## Expected Outcomes

#

- Comparison of migration kinetics for both additives.
- Understanding of additive behavior in food packaging.
- Ready-to-use figures and reports for food safety analysis.

## Dependencies

This script relies on several **SFPPy** modules:

| Module                                       | Purpose                                                                                                                                             |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>patankar.loadpubchem</code>            | Retrieves chemical properties of migrants from PubChem.          |
| <code>patankar.geometry</code>               | Defines the 3D geometry of the packaging.                        |
| <code>patankar.food</code>                   | Models the food layer and its interaction with packaging.        |
| <code>patankar.layer</code>                  | Provides polymer layer properties and diffusion parameters.    |
| <code>patankar.migration</code>              | Simulates mass transfer using a <b>finite-volume solver</b> .  |
| <code>patankar.migration.print_figure</code> | Exports and saves migration results.                           |

## Code Breakdown

### 1 Define Contact Conditions

#

```
contactTemperature = (7, "degC") # Temperature of the sandwich
contactTime = (10, "days") # Contact duration
```

### 2 Define Sandwich Geometry

#

```
sandwich_geom = Packaging3D(
    'Cylinder',
    height=(19, "cm"), # 19 cm height (length=height)
    radius=(6, "mm") # 6 mm radius
)
internalvolume, contactsurface = sandwich_geom.get_volume_and_area()
```

### 3 Define the Migrant ( Irganox 1076 )

#

```
m1 = migrant("irganox 1076")
```

### 4 Create the LDPE Film

#

```
LDPElayer_with_m1 = polymer.LDPE(  
    l=(100, "um"), # Thickness: 100 µm  
    substance=m1,  
    C0=5000, # Initial concentration (arbitrary units, but let's assume 5000 mg/kg)  
    T=contactTemperature  
)
```

### 5 Define Food Properties

#

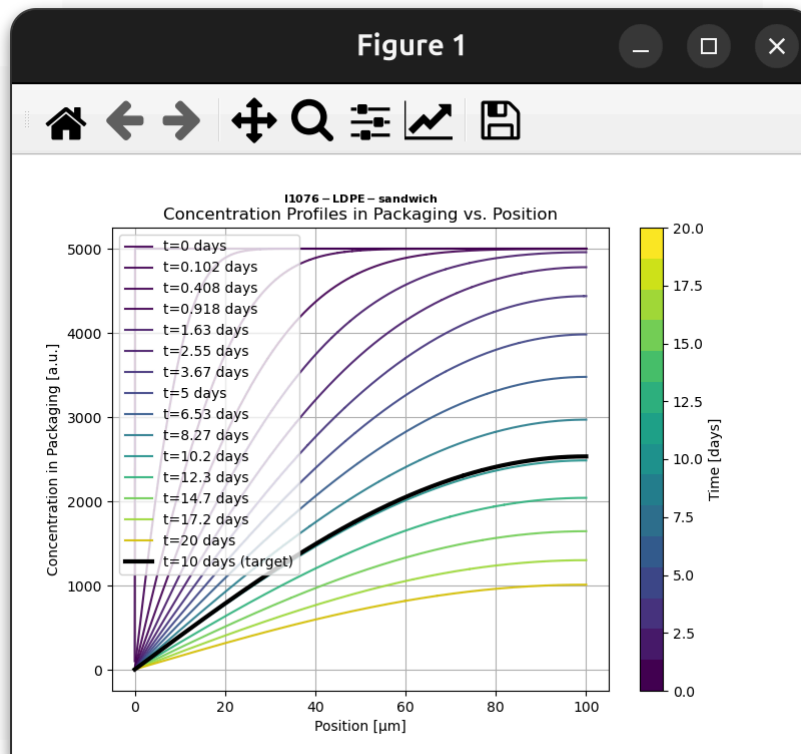
```
class sandwich(food.realfood, food.semisolid, food.fat):  
    name = "sandwich"  
  
FOODlayer = sandwich(  
    volume=internalvolume,  
    surfacearea=contactsurface,  
    contacttime=contactTime,  
    contacttemperature=contactTemperature,  
    substance=m1, # Required for partition coefficient prediction  
    simulant="ethanol" # a simulant is required to predict partition coefficients  
)
```

### 6 Run Migration Simulation

#

```
simulation = solver(  
    LDPElayer_with_m1, # LDPE containing Irganox 1076  
    FOODlayer, # Fatty sandwich (here with m1)  
    name="I1076-LDPE-sandwich"  
) # run the simulation  
  
# Display simulation details  
print(simulation)  
# Retrieve concentration at target time in SI units  
print("CF at time t=", simulation.ttarget, "[s] = ", simulation.CFtarget, "[a.u.]")  
  
# Interpolate Concentrations at Specific Times  
tnew = _toSI([3, 8, 9, 10, 12, 14, 18], "days").flatten()  
CF = simulation.interpolate_CF(tnew).flatten()  
  
# Concentration profiles at ttarget  
simulation.plotCx()
```

```
<SensPatankarResult:I1076-LDPE-sandwich: CF(10 [days]) = 116.4 [a.u.]> CF at time t= [864000] [s] = [116.40897281] [a.u.]
```



## 7 Compare Migration for Two Additives: Irganox 1076 (m1) and Irgafos 168 (m2) #

```
# define m2
m2 = migrant("irgafos 168")
# update the formulation of the film
LDPElayer_with_m2 = polymer.LDPE(
    l=(100, "um"),
    substance=m2,
    C0=5000,
    T=contactTemperature
)
simulation2 = solver(
    LDPElayer_with_m2,          # contact material
    FOODlayer.update(substance=m2), # contact medium
    name="I168-LDPE-sandwich"    # simulation name
) # run for m2

# Add the new simulation (Irgafos 168) to the results store
allCF.add(simulation2, "Irgafos 168", "b") # Assign blue color "b"

# Generate the corresponding plot
hfig2 = simulation2.plotCF(t=tnew) # hfig2 = figure handle

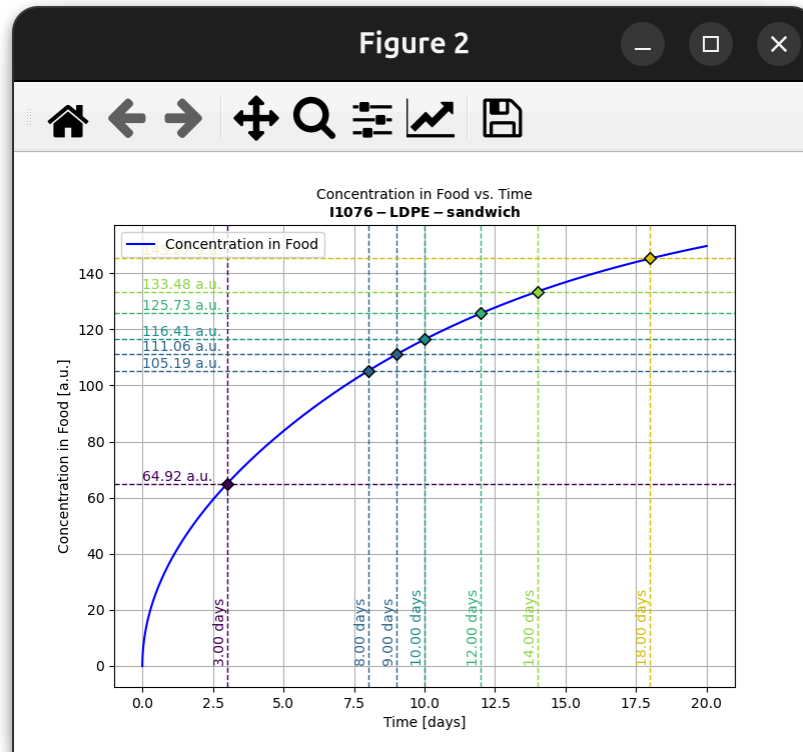
# Compare CF vs time for Irganox 1076 (m1) and Irgafos 168 (m2)
hfig12 = allCF.plotCF() # hfig12 figure handle for the comparison
```

## 8 Save and Print Figures

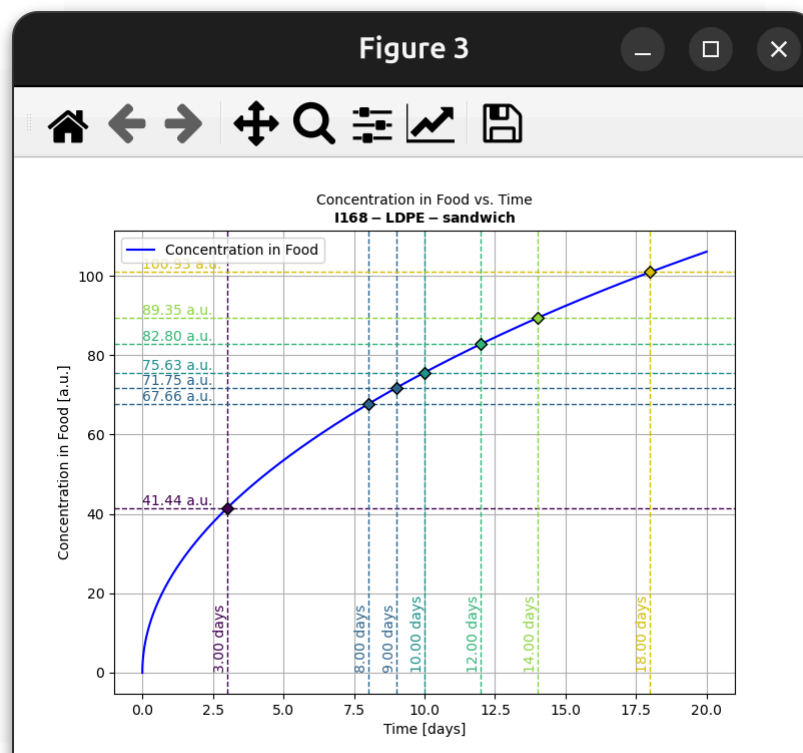
#

```
print_figure(simulation.plotCF(), **printconfig) # plot and print
print_figure(simulation2.plotCF(), **printconfig) # plot and print
```

For Irganox 1076 :



For Irgafos 168 :



 Estimated partition coefficients:

💡 Since partition coefficients are estimated in a conservative manner, substances with similar molar volume and polarity are likely to be represented with similar partition coefficients. *Food-to-packaging* partition coefficient reads:

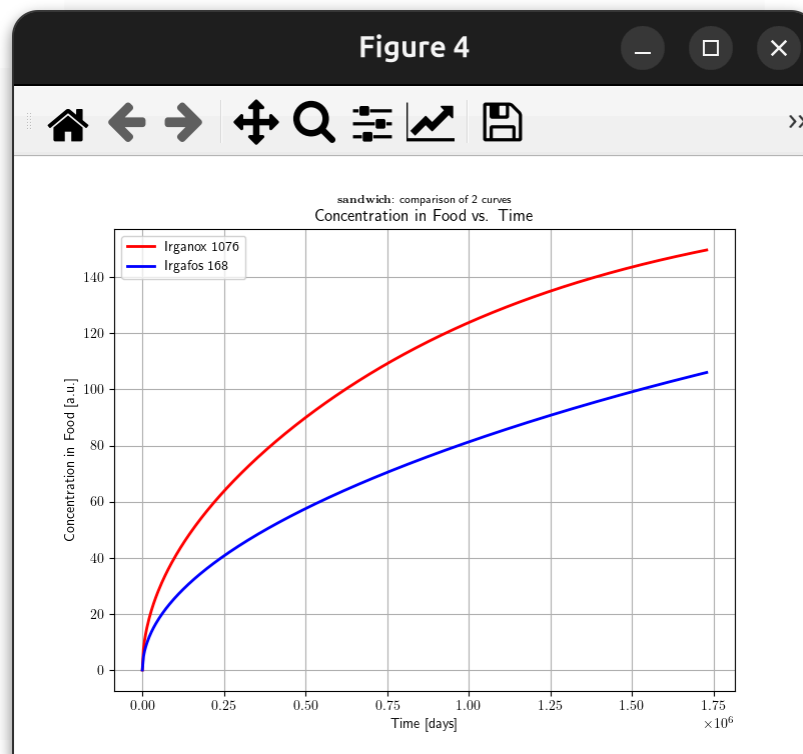
$$C_{F/P} = \frac{C_F^{eq}}{C_P^{eq}} = \frac{k}{k_0} \quad (1)$$

```
KFP_m1 = LDPElayer_with_m1.k / FOODlayer.update(substance=m1).k0
KFP_m2 = LDPElayer_with_m2.k / FOODlayer.update(substance=m2).k0
print(f"m1:I1076 KF/P = {KFP_m1}", "\n", f"m2:I168 KF/P = {KFP_m2}")
```

m1:I1076 KF/P = [24.5896934] m2:I168 KF/P = [24.5896934]

 Kinetic Comparison

```
allCF = store(name="sandwich")
allCF.add(simulation, "Irganox 1076", "r") # Assign red color "r" to I1076
allCF.add(simulation2, "Irgafos 168", "b") # Assign blue color "b" to I168
hfig12 = allCF.plotCF() # hfig12 figure handle for the comparison
```





# SFPPy for Food Contact Compliance and Risk Assessment

Contact [Olivier Vitrac](#) for questions | [Website](#) | [Documentation](#)